ITMO Machine Learning Technologies – September–November 2026



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PhD

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Topics:

- classic machine learning algorithms and data analysis
- introduction to neural networks and deep learning

- **Telegram group:**

Schedule:

- **Lectures:** at 19-00 (online) on Mondays
- **Labs:** at 20-30 (online) on Mondays
- **Exam:** December 1st 2026
- **Course project:** (?)

Course grade format:

- There will be HW assignments in a form of practical assignments + online tests
- 3 (any of 4) passed HW assignments + 2 (any of 3) passed online tests is your permission to exam;
- **"4C" automatically** if you have all 4 tasks and 3 tests in time (deadline is ~25.11)

Plan of the course:

Lectures:

1. Introduction to ML
2. Data preprocessing and algorithms overview
3. Unsupervised Learning
4. Natural Language processing
5. Optimization methods
6. MLP and backpropagation algorithm
7. Computer vision and CNNs
8. Sequence analysis and prediction

Practical classes
(Jupyter Notebooks examples and discussion):

1. Python for data science (overview)
2. Classification & regression algorithms
3. Classification metrics – **task #1**
4. Natural Language processing – **task #2**
5. Clustering algorithms
6. Optimization methods
7. Backpropagation algorithm – **task #3**
8. ML task solving general pipeline
9. CNNs – **task #4** ***Contest!***
10. Time series analysis and prediction
11. Dimension reduction and autoencoders
12. Introduction to generative models

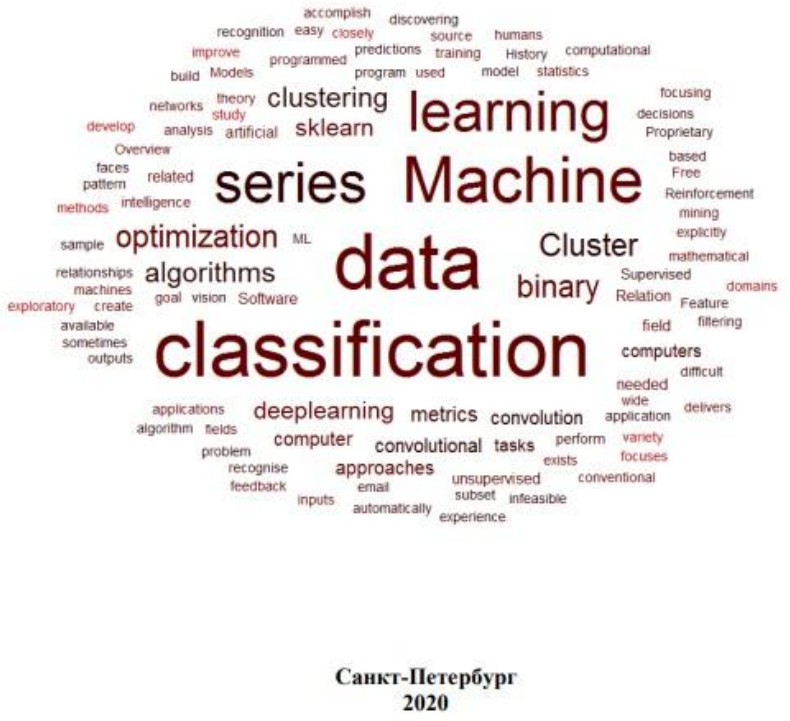
Important Information about HW tasks:

1. Should be sent **before deadline (~25.11.2026).** Deadline is **one for all the tasks.**
2. You can send it in a form of **google colab or github link.**
3. Send directly to @petrgl via Telegram
4. Template for the name of online notebooks: **“Task_{#}_{Name}_Group_{##}.xxx”.**
5. You may submit assignments after the deadline; however, in that case, you will be required to take the exam—you forfeit the right to have your exam grade automatically awarded based on your coursework.
6. Your progress will be shown in a shared **Google Sheet**

   (to be shared further in **MLT 2026** group)

Textbook (in Russian)
In part can be used for preparation of hometasks

https://books.ifmo.ru/file/pdf/2633.pdf



Small introductory test (not graded):

https://forms.gle/kCK4FTYuEWdHdpbj8